

Lecture 6 - Support Vector Machines

Outline

- Naive Bayes
- Laplace Smoothing
- Event models for text classification

Common advice on applying ML

- SVMs intro.

For Naive Bayes,

$$P(y=1|x) = \frac{P(x|y=1) P(y=1)}{P(x|y=1) P(y=1) + P(x|y=0) P(y=0)}$$

by Baye's rule

NIPS conference paper →

$$P(y=1|x) = \frac{P(x|y=1) P(y=1)}{P(x|y=1) P(y=1) + P(x|y=0) P(y=0)}$$

$$j = 6017$$

$$P(x_{6017=1} | y=1) = \frac{0}{\text{\# no. of examples}}$$

$$P(x_{6017=1} | y=0) = \frac{0}{\text{\# no. of examples}}$$

so

probabilistically, its $\frac{0}{0+0}$.

team

If a football team has lost 4 games consecutively, the probability of winning the 5th game is not 0 (statistically incorrect).

$$p(y=1) = \frac{\# \text{ victories} + 1}{\# \text{ losses} + 1 + \# \text{ victories} + 1}$$

$$\# \text{ losses} + 1 + \# \text{ victories} + 1$$

$$\# \text{ losses} + 1 + \# \text{ victories} + 1$$

$$= \frac{1}{6} \text{ by Laplace Smoothing}$$

So, building a spam classifier

In general, $X \in \{1, \dots, K\}$

Estimate

$$P(X=i) = \frac{\sum_{i=1}^m \mathbb{1}\{x^{(i)} = i\} + 1}{m + K}$$

For spam classifier

$$\phi_{j|y=0} = \frac{\sum_{i=1}^m \mathbb{1}\{x_j^{(i)} = 1 \wedge y^{(i)} = 0\} + 1}{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 0\} + 2}$$

$$\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 0\} + 2$$

consider a classification problem

$X_i \in \{1, 2, \dots, k\}$

size	< 400	$400 - 800$	$800 - 1200$	> 1200
X_i	1	2	3	4

The problem is what is the probability it will be sold in the next 30 days not

$$P(X|y) = \prod_{i=1}^n P(X_i|y)$$

multinomial probability
it can be
probability mass
function

Multi variate Bernoulli event model

$X =$	0	$\rightarrow$	no $\leftarrow 1$
	0	$\rightarrow$	adrank $\leftarrow 2$
	1	$\rightarrow$	buy $\leftarrow 800$
	1	$\rightarrow$	drug $\leftarrow 1600$
	1	$\rightarrow$	now $\leftarrow 6200$

for the email, "Drugs, Buy drugs now"

Multinomial event model.

$$x \in \begin{bmatrix} 1600 \\ 800 \\ 1600 \\ 620 \end{bmatrix} \in \mathbb{R}^n$$

$$x_j \in \{1, \dots, 10,000\}$$

$n_j =$ ~~length~~ length of email

$$P(x, y) = P(x|y) \cdot P(y)$$

$$\text{assume} = \prod_{j=1}^n P(x_j|y) \cdot P(y)$$

↓
multinomial
probability.

Parameters $\phi_y = P(y=1)$

$$\phi_{k|y=0} = P(x_j = k | y=0)$$

↓

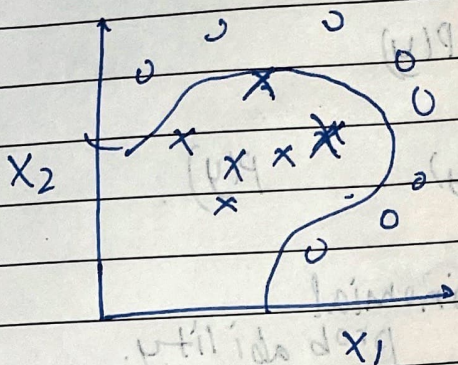
chance of word
being k if $y=0$.

$$\phi_{k|y=1} = P(x_j = k | y=1)$$

by applying MLE:

$$\phi_{K|y=0} = \frac{\sum_{i=1}^m 1\{y^{(i)}=0\} \sum_{j=1}^{n_i} x_j^{(i)} + \sum_{i=1}^m 1\{y^{(i)}=1\} \sum_{j=1}^{n_i} x_j^{(i)}}{\sum_{i=1}^m 1\{y^{(i)}=0\} n_i + 10000}$$

Support Vector Machines



For non linear decision boundaries, we use SVM.

- optimal margin classifier (separable class).

- Kernels

- Inseparable class

Functional margin $\Rightarrow$

$h(x) = g(\theta^T x)$ - Logistic classifier.

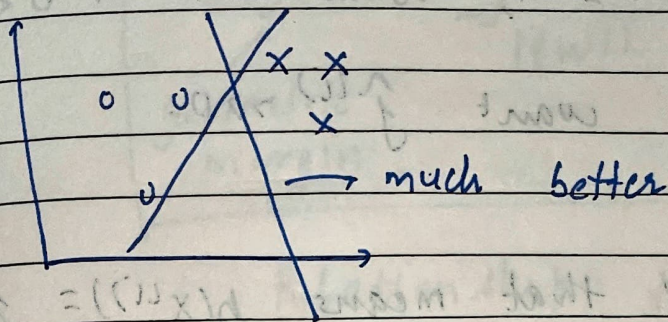
Predict "1" if $\theta^T x \geq 0$
Predict "0" otherwise

$$z = \theta^T x \rightarrow g(\theta^T x) = \frac{1}{1 + e^{-z}}$$

If $y^{(i)} = 1$, hope that $\theta^T x^{(i)} \gg 0$ ^{much greater}

If $y^{(i)} = 0$, hope that $\theta^T x^{(i)} \ll 0$

Geometric Margin



Notation -

Labels $y \in \{-1, 1\}$
 Have n outputs values in $\{-1, 1\}$

$$g(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

Previously $\rightarrow h_{\theta}(x) = g(\theta^T x)$
 $\uparrow \mathbb{R}^{n+1}$

$$h_{w,b}(x) = g(w^T x + b)$$

$\uparrow \mathbb{R}$
 $\sum_{i=1}^n w_i x_i + b$
 $\uparrow \mathbb{R}^n$

Drop $x_0 = 1$
 notation

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} \begin{matrix} y_b \\ \\ \\ \end{matrix} \} w$$

Functional margin of hyperplane defined by (w, b) wrt $(x^{(i)}, y^{(i)})$

$$\hat{f}^{(i)} = y^{(i)} (w^T x^{(i)} + b)$$

If $y^{(i)} = 1$, want $w^T x^{(i)} + b \gg 0$

If $y^{(i)} = -1$, want $w^T x^{(i)} + b \ll 0$

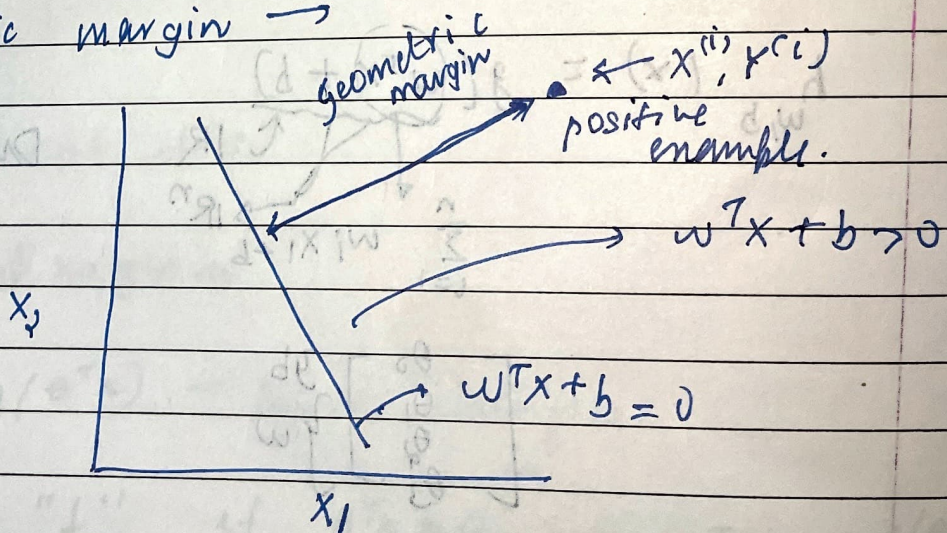
so we want $\hat{f}^{(i)} \gg 0$.

If $\hat{f}^{(i)} > 0$, that means $h(x^{(i)}) = y^{(i)}$
 functional margin wrt training set

$$\hat{f} = \min_{i=1, \dots, m} \hat{f}^{(i)}$$

$$(w, b) \rightarrow \left(\frac{w}{\|w\|}, \frac{b}{\|b\|} \right)$$

Geometric margin $\rightarrow$



Geometric margin of hyperplane (w, b)
 wrt. $x^{(i)}, y^{(i)}$

$$j^{(i)} = \frac{y^{(i)} (w^T x^{(i)} + b)}{\|w\|}$$

$$\boxed{\begin{array}{l} j^{(i)} = \frac{\hat{j}^{(i)}}{\|w\|} \rightarrow \text{functional margin.} \\ \uparrow \\ \text{geometric margin} \end{array}}$$

Relationship between
 geometric margin & functional margin.

Geometric margin wrt training set

$$j^{(i)} = \min_i j^{(i)}$$

Optimal margin classifier

choose w, b to maximize $j^{(i)} \rightarrow \text{gm}$

max.

$$j, w, b \text{ subject to } \frac{y^{(i)} (w^T x^{(i)} + b)}{\|w\|} \geq j \quad i=1, \dots, m$$

min
 w, b

$$\downarrow \|w\|^2$$

$$\text{subject to } \frac{y^{(i)} (w^T x^{(i)} + b)}{\|w\|^2} \geq 1$$

$$y^{(i)} (w^T x^{(i)} + b) \geq 1$$